How Waiting Time Shapes Preventive Health Behavior: Evidence from Vaccination Decisions

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Abstract

Timely vaccination is critical for reducing infectious disease risk, yet many individuals delay immunization even when vaccines are available. This study examines waiting time as a behavioral barrier to preventive health decision-making and evaluates whether institutional trust and medical vulnerability modify responses to delayed protection. We analyzed data from a discrete choice experiment conducted among 1,027 adults in Wuhan, China. Respondents evaluated hypothetical vaccines that varied in efficacy, side effects, waiting time, and monetary incentives. Mixed logit models were used to estimate the behavioral impact of delay and to assess heterogeneity across subgroups.

Longer waiting times significantly reduced the likelihood of vaccine uptake, indicating that delayed protection carries a meaningful behavioral cost. Individuals with lower institutional trust were substantially more sensitive to delay than those with higher trust. Medically vulnerable respondents also exhibited greater aversion to waiting, suggesting that the burden of delay is not evenly distributed across the population. Smokers appeared relatively more tolerant of delayed protection, highlighting additional heterogeneity in preventive decision-making. In an exploratory extension, these behavioral parameters were incorporated into an epidemic simulation to illustrate possible population-level implications of delayed uptake.

These findings suggest that waiting time is an important behavioral determinant of vaccination decisions. Reducing administrative delay and strengthening institutional trust may improve the timeliness and equity of preventive health delivery.

Keywords: vaccination, waiting time, preventive health behavior, institutional trust, medical vulnerability, delayed protection

1. Introduction

Timely vaccination is central to reducing exposure to infectious disease and protecting health. Although vaccine uptake is often treated as a binary outcome, the timing of uptake also matters. When vaccination is postponed, individuals remain unprotected for longer periods, which may extend exposure to infection risk and increase the likelihood of adverse health consequences [1, 2, 3]. Vaccine delay is therefore not only a logistical issue but also a behavioral pattern with important implications for preventive health.

From a behavioral perspective, vaccination involves a trade-off between immediate burdens and future protection. Individuals may weigh the effort of obtaining a vaccine, inconvenience, concerns about side effects, or other near-term costs against the delayed benefit of reduced infection risk [4]. These trade-offs are unlikely to be uniform across the population. Institutional trust [5], perceived medical vulnerability [6], and preventive-health routines [7] may shape how individuals respond to delayed protection, creating heterogeneity in tolerance for waiting and in the likelihood of vaccine delay.

A growing literature suggests that vaccination behavior is influenced by impatience, risk perception, institutional trust, and access-related barriers [8, 9, 10]. However, waiting time is often treated primarily as an administrative friction rather than as a behavioral determinant of preventive decision-making [11, 12, 13]. This distinction matters. As an experimentally manipulable service attribute, waiting time may directly affect willingness to vaccinate by altering how individuals evaluate delayed protection. Yet rela-

tively few discrete choice studies have directly examined waiting time in this way, and fewer still have linked sensitivity to waiting time to institutional trust or chronic medical vulnerability [14, 15, 16].

To address these gaps, this study examines waiting time as a behavioral barrier to vaccination. We analyze data from a discrete choice experiment administered to 1,027 adults in Wuhan, China, in which waiting time was embedded directly as a vaccine attribute. This design allows us to estimate how delayed protection influences vaccination choices and whether these responses differ across population subgroups. Rather than treating waiting time as a purely operational detail, we examine it as a feature of preventive health delivery that may shape decision-making under uncertainty.

Wuhan provides an informative setting for this analysis because of its high population density, early pandemic experience, and strong public-health visibility. In such a context, delayed protection is likely to be salient to respondents, making it possible to observe how trust, perceived vulnerability, and health-related routines shape responses to waiting. Although the magnitude of these responses may vary across settings, the underlying processes are likely to be relevant more broadly, particularly in contexts where institutional trust plays an important role in preventive health behavior [17, 18].

This study makes three contributions. First, it provides evidence that waiting time is an important behavioral determinant of vaccination decisions. Second, it shows that sensitivity to waiting time is patterned by institutional trust, chronic medical conditions, and preventive-health routines, indicat-

ing that responses to delayed protection vary systematically across groups. Third, by translating these differences into measures of perceived delay burden, the analysis highlights how the same waiting period may be experienced differently across the population. In an exploratory extension, we also examine how these behavioral responses may relate to population-level epidemic outcomes.

Together, these findings suggest that waiting time is not merely a logistical feature of vaccine delivery but a meaningful behavioral factor in preventive health decision-making. Understanding how waiting time contributes to vaccine delay may help inform more timely and equitable vaccination strategies.

2. Methods

This study used a discrete choice experiment (DCE) to examine how waiting time influences vaccination decisions and whether responses to delayed protection differ across population subgroups. Waiting time was embedded directly as a vaccine attribute, allowing us to estimate how respondents traded off delayed protection against other vaccine characteristics. In addition to estimating preference heterogeneity, we derived summary measures of time-related valuation from the choice-model coefficients. The main text focuses on the behavioral interpretation of these measures, while additional technical details and validity checks are reported in the online supplementary material.

2.1. Study Design

This study used an online discrete choice experiment (DCE) to examine how **waiting time** influences vaccination decisions and whether responses to delayed protection differ across population subgroups. Respondents evaluated hypothetical vaccines for a COVID-like disease (CLD), defined as a respiratory infection with transmissibility and severity comparable to COVID-19. This framing was intended to preserve a realistic infectious-disease context while reducing anchoring to a specific epidemic wave, variant, or policy period.

The CLD scenario was used to elicit preferences for delayed protection without tying responses to a single historical phase of the COVID-19 pandemic. Similar stated-preference studies have used hypothetical vaccine scenarios to examine vaccination preferences and timing decisions [19, 20, 21]. The survey was administered online in early 2025, when vaccination services were broadly available and respondents were no longer making decisions under the immediate pressures of an acute outbreak.

2.2. Participants and Setting

The survey targeted adults living in Wuhan, China. Sampling quotas for age, gender, and district were used to approximate the demographic composition of the adult population. Eligibility criteria required respondents to be aged 18 years or older, currently reside in Wuhan, and complete all survey modules. Standard attention and data-quality checks were applied

before analysis. The final analytic sample consisted of 1,027 respondents. Additional information on sample composition and comparison with Wuhan Census benchmarks is provided in the online supplementary material.

Wuhan provides an informative setting for studying responses to delayed protection because of its early pandemic experience, strong public-health visibility, and high salience of vaccination-related decisions. These features make it possible to examine how institutional trust, perceived vulnerability, and preventive-health routines may shape responses to **waiting time** in a context where infectious disease risk is readily understood.

2.3. DCE Attributes and Choice Tasks

Each vaccine profile was defined by five attributes: **waiting time** (0–6 months), vaccine efficacy, expected side effects, vaccine origin, and cash incentives. These attributes were selected to capture key features of vaccination decisions while allowing respondents to evaluate trade-offs between delayed protection and other vaccine characteristics.

Each respondent completed six choice tasks. In each task, respondents chose between two hypothetical vaccine alternatives or an opt-out option. An efficient fractional-factorial design was used to achieve attribute balance and support identification of main effects.

Prior to the main survey, the instrument was pilot-tested with 60 respondents to assess clarity, comprehension, and cognitive burden. Pilot responses informed the priors used in the efficient design algorithm and indicated that

task complexity was manageable for the target population. Additional details on attribute definitions, example choice tasks, pilot-based instrument development, and design diagnostics are provided in the online supplementary material.

2.4. Statistical Analysis

We estimated mixed logit models to account for repeated choices and unobserved heterogeneity in vaccination preferences. These models were used to estimate the behavioral impact of **waiting time** on vaccine choice and to assess whether sensitivity to delayed protection differed across subgroups defined by demographic characteristics, health behaviors, chronic medical conditions, and institutional trust.

To summarize time-related valuation, we also derived exponential discounting (δ), quasi-hyperbolic present bias (κ), and marginal willingness-to-accept (MWTA) for **waiting time** from the estimated choice-model coefficients. These measures were used as summary indicators of the perceived burden of delayed protection and to compare responses across subgroups.

All analyses were conducted in R using the `mlogit` and `gmnl` packages. Full model specifications, formal derivations, robustness analyses, epidemic-model details, and supplementary figures are provided in the online supplementary material.

3. Results

3.1. Main Model Estimates

Table 1 presents estimates from the conditional logit, multinomial logit, and mixed logit models. To facilitate comparison across attributes, coefficients were standardized to represent the change in utility associated with a one-standard-deviation change in each attribute.

Across all models, **waiting time** was negatively associated with vaccine uptake, indicating that longer waits substantially reduced willingness to vaccinate. Respondents showed positive preferences for vaccine efficacy and cash incentives, whereas side effects had weaker and less consistent associations with uptake. In substantive terms, the disutility associated with longer waiting time was partly offset by increases in cash incentives and vaccine efficacy, suggesting that respondents traded off time, money, and expected protection when making vaccination decisions.

The mixed logit model provided the best fit based on the log-likelihood and Akaike Information Criterion (AIC). Its random-parameter specification also captured substantial heterogeneity in responses to **waiting time**, as reflected in the significant standard deviation of the waiting-time parameter. We therefore use the mixed logit estimates as the primary basis for subsequent analyses of discounting, subgroup variation, and perceived delay burden.

Table 1 shows the comparison of logit models.

Table 1: Comparison of logit models (dependent variable: choice)

	Conditional logit	MNL (ASCs)	Mixed logit
Waiting time (std)	-0.131^{***} (0.019)	-0.125^{***} (0.020)	-0.353^{***} (0.027)
Vaccine efficacy (std)	0.153^{***} (0.036)	0.163^{***} (0.037)	0.218^{***} (0.041)
Side effects (std)	-0.042 (0.027)	-0.082^{***} (0.030)	0.008 (0.031)
Cash incentives (std)	0.327^{***} (0.018)	0.313^{***} (0.019)	0.431^{***} (0.025)
Vaccine origin (imported)	0.244^{***} (0.038)	0.239^{***} (0.038)	0.093^{**} (0.047)
Opt-out ASC		-0.214^{**} (0.093)	-0.287^{**} (0.118)
ASC: B		-0.108^{***} (0.033)	
ASC: C		-0.326^{***} (0.100)	
SD waiting time (std)			1.303^{***} (0.073)
Num. obs.	6162	6162	6162
Log likelihood	-6101.771	-6096.612	-5615.522
AIC	12215.543	12207.225	11245.044

Note: Standard errors in parentheses. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

3.2. Discounting Profiles

Figure 1 presents the estimated discounting profiles from the exponential and hyperbolic models fitted to the observed valuation points. Both curves begin at full value at zero delay and decline monotonically as **waiting time** increases. The exponential model yielded an estimated discount factor of $\delta = 0.160$, whereas the hyperbolic model yielded $\kappa = 0.225$. The hyperbolic curve showed a sharper initial decline over the first several months, indicating stronger short-run sensitivity to delayed protection.

These estimates suggest that respondents placed disproportionate weight on immediacy when evaluating vaccination. In other words, the behavioral cost of **waiting time** was not linear across the delay horizon: shorter delays near the present carried especially large behavioral consequences relative to

equal delays further in the future. Additional subgroup-specific discounting estimates are reported in the online supplementary material.

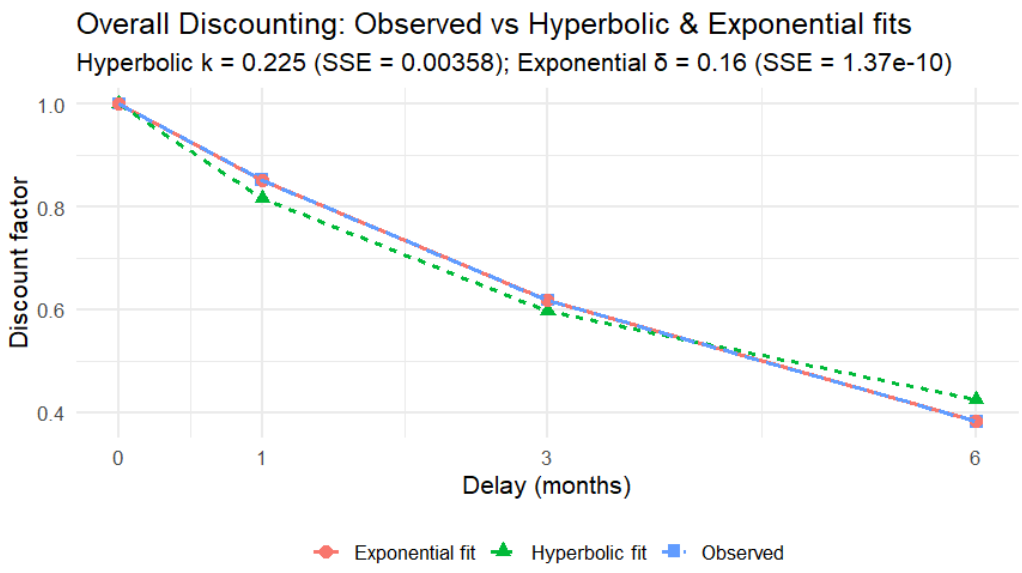

Figure 1: Comparison of observed discounting with hyperbolic and exponential fits.

Alt text: A line chart comparing observed discounting behaviour with fitted hyperbolic and exponential curves, showing close alignment in early months and divergence over longer delays.

3.3. Subgroup Variation in Responses to Waiting Time

Responses to **waiting time** varied meaningfully across demographic, behavioral, health-related, and institutional subgroups. Figure 2 summarizes variation in the hyperbolic discount parameter (κ), where higher values indicate stronger present-oriented preferences and greater sensitivity to delayed protection.

Several patterns were especially notable. Respondents with lower institutional trust exhibited substantially greater sensitivity to **waiting time** than those with higher trust. Individuals reporting chronic medical conditions also showed steeper discounting, indicating that delayed protection may be perceived as especially costly among medically vulnerable groups. Variation was also observed across age, education, smoking status, and preventive-health routines, suggesting that the behavioral burden of waiting is not evenly distributed across the population.

Detailed subgroup estimates for both hyperbolic and exponential discounting are reported in the online supplementary material.

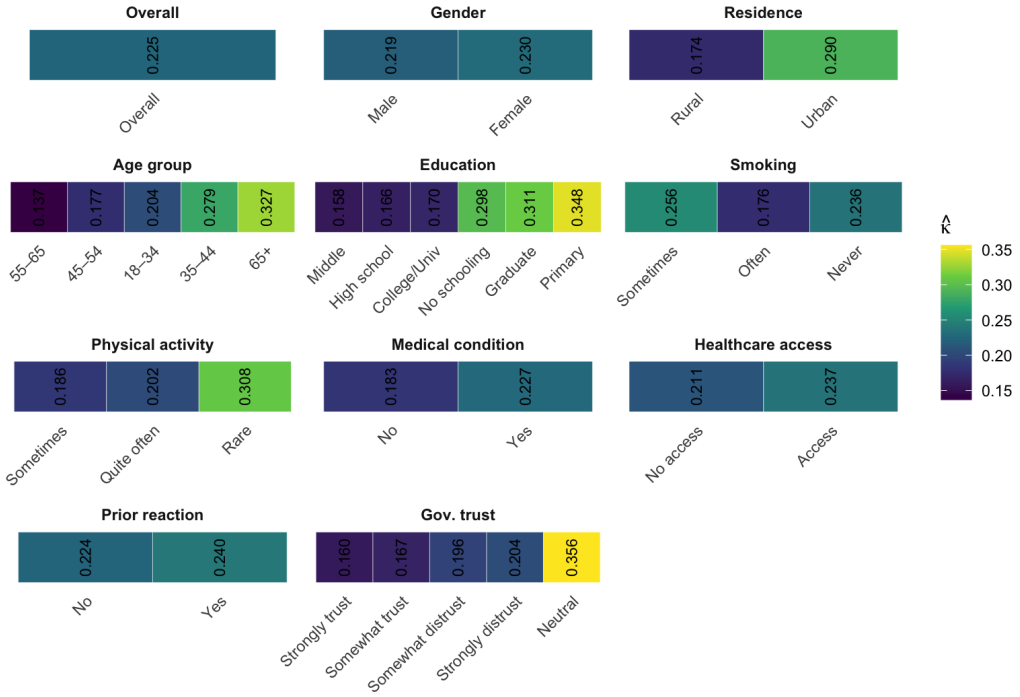

Figure 2: Subgroup variation in hyperbolic discount parameters (κ). Higher values indicate stronger present-oriented preferences (greater impatience toward waiting time). Clear gradients appear across age, education, health behavior, medical vulnerability, and institutional trust.

3.4. Perceived Delay Burden

We further summarized the behavioral cost of **waiting time** using marginal willingness-to-accept (MWTA), which expresses the compensation respondents would require to tolerate an additional month of waiting. Across the full sample, respondents required approximately 45–56 RMB per month of delay, indicating that **waiting time** imposed a meaningful perceived burden on vaccination decisions.

MWTA estimates also revealed substantial heterogeneity across subgroups.

Respondents with lower institutional trust, lower education, chronic medical conditions, or less regular preventive-health routines generally required greater compensation to accept additional waiting, consistent with the subgroup differences observed in the discounting parameters. For example, respondents with low institutional trust required considerably more compensation for delay than those with high trust. These patterns suggest that the same **waiting time** may be experienced quite differently across the population.

For brevity, subgroup-specific MWTA estimates and related policy-translation figures are reported in the online supplementary material.

3.5. Exploratory Population-Level Extension

As an exploratory extension, we incorporated the estimated behavioral responses to **waiting time** into a behaviorally augmented SEIR framework to illustrate possible population-level implications of delayed uptake. In these simulations, behavioral sensitivity to **waiting time** reduced the effective pace of vaccination relative to a counterfactual scenario driven only by logistical capacity.

The simulation results suggest that even moderate delay aversion may alter epidemic trajectories by shifting uptake later in the outbreak curve, when early protection would otherwise have greater epidemiological value. In the illustrative scenarios considered here, behavioral responses to delayed protection were associated with higher cumulative attack rates than the supply-only

benchmark, consistent with later effective uptake under stronger sensitivity to waiting time. This pattern was more pronounced in scenarios representing subgroups with greater sensitivity to **waiting time**. Full model equations, calibration procedures, and extended simulation results are provided in the online supplementary material.

4. Discussion

This study examined how **waiting time** influences vaccination decisions in a hypothetical COVID-like disease setting and whether responses to delayed protection differ across population subgroups. Three main findings emerged. First, longer **waiting time** substantially reduced willingness to vaccinate, indicating that delayed protection functions as a meaningful behavioral barrier in preventive health decision-making. Second, responses to **waiting time** were not uniform across the population. Individuals with lower institutional trust and those reporting chronic medical conditions were more sensitive to delayed protection, suggesting that the burden of waiting is shaped by both social and health-related vulnerability. Third, these behavioral patterns were also reflected in discounting-based and monetary summary measures, indicating that the same period of **waiting time** may be experienced differently across groups.

4.1. Waiting Time as a Behavioral Barrier to Vaccination

The present findings add to a growing literature showing that vaccination decisions are shaped not only by perceived efficacy and side effects, but also

by how individuals evaluate the timing of protection [16, 22, 23]. In this study, **waiting time** was embedded directly as an experimentally manipulated vaccine attribute, allowing us to estimate how delayed protection affects stated vaccination choices. The results suggest that even relatively short delays can carry substantial behavioral weight, consistent with evidence that near-term frictions often have disproportionate effects on preventive health behavior [24, 21, 25].

This interpretation helps clarify why *vaccine delay* persists even when vaccines are available. Delayed access is not experienced merely as an administrative inconvenience. Rather, **waiting time** appears to influence how respondents weigh immediate burdens against future protection, especially under uncertainty. In this sense, the results are consistent with prior work showing that present-oriented decision-making can shape vaccination timing and other preventive choices [26, 27, 11, 12]. At the same time, the current findings suggest that these behavioral responses are closely tied to the context of health protection, rather than reflecting a single, context-free trait.

4.2. Institutional Trust and Medical Vulnerability

Institutional trust emerged as an especially important moderator of responses to **waiting time**. Respondents with lower trust placed a substantially greater burden on delayed protection than those with higher trust. This pattern is consistent with prior research linking trust to vaccine acceptance, health communication, and adherence to public-health recommenda-

tions [24, 28]. When trust is low, delayed access may be interpreted not only as inconvenience, but also as a signal of uncertainty about whether promised protection will be delivered effectively or at all. Under these conditions, the behavioral impact of **waiting time** may be amplified, increasing the likelihood of *vaccine delay* among populations that are already more skeptical of institutions or less confident in public-health systems.

Medical vulnerability showed a similar pattern. Respondents reporting chronic medical conditions were more sensitive to delayed protection, suggesting that **waiting time** may carry greater perceived cost among those facing elevated health risk. This finding is consistent with the view that responses to delayed vaccination are shaped by perceived susceptibility as well as by general behavioral tendencies [6, 24]. More broadly, it suggests that heterogeneity in vaccination timing is partly structured by differences in health status, trust, and preventive-health routines. These results therefore support interpretations of *vaccine delay* as socially patterned and health-relevant, rather than simply random variation in patience or convenience preferences.

Variation was also observed across age, education, smoking status, and physical activity. These subgroup patterns should be interpreted cautiously, but they reinforce the broader conclusion that the behavioral impact of **waiting time** is not evenly distributed across the population. Additional subgroup-specific estimates are reported in the online supplementary material.

4.3. Implications for Preventive Health Delivery

These findings have several implications for preventive health practice. First, they suggest that reducing **waiting time** may improve uptake even when vaccines are widely available, because service delays can function as behavioral barriers rather than merely logistical frictions. Second, they indicate that the consequences of **waiting time** may be greater for groups with lower institutional trust and greater medical vulnerability. This suggests that clear communication, predictable scheduling, and timely access may be especially important in populations that are more sensitive to delayed protection. Third, the results highlight the value of considering behavioral responses to service delivery when designing vaccination programs, particularly in settings where timely uptake is critical.

The monetary summary measures reported in this study are useful for interpretation because they translate the burden of **waiting time** into a more concrete metric. However, the central implication is behavioral rather than purely economic: the same delay may carry very different meanings across groups. From a public-health perspective, this reinforces the importance of service design, including how quickly protection can be delivered and how clearly the process is communicated.

The exploratory epidemic simulations provide an additional illustration of why these behavioral responses may matter beyond the individual level. When incorporated into a simplified epidemic model, sensitivity to **waiting time** was associated with slower effective uptake and less favorable outbreak

dynamics. These simulations should be interpreted as illustrative rather than definitive, but they suggest that micro-level responses to delayed protection may have broader public-health consequences. Full model details and calibration procedures are provided in the online supplementary material.

4.4. Limitations

Several limitations should be noted. First, as with all stated-preference studies, hypothetical bias may affect the absolute magnitude of estimated valuations. Although the survey design incorporated realistic attribute ranges, repeated choice tasks, an opt-out option, and pilot-based refinement, the results should be interpreted primarily as evidence on relative behavioral patterns rather than exact population-level parameters [19, 21, 20]. Second, the study was conducted in Wuhan, a setting with unusually high salience of infectious disease and vaccination, which may influence the magnitude of observed responses. Future work should examine whether similar patterns appear in lower-salience or non-pandemic settings. Third, chronic medical conditions were self-reported rather than clinically verified, and future studies could strengthen interpretation by incorporating objective health measures. Finally, the epidemic-modeling component is exploratory and should not be interpreted as a full policy evaluation.

Taken together, these findings suggest that **waiting time** is an important behavioral determinant of vaccination decisions and that its effects are amplified by differences in institutional trust and medical vulnerability. Un-

derstanding how **waiting time** contributes to *vaccine delay* may help inform more timely and equitable preventive health delivery.

5. Conclusion

This study shows that **waiting time** is an important behavioral determinant of vaccination decisions. Longer waits reduced willingness to vaccinate, and this effect was stronger among respondents with lower institutional trust and greater medical vulnerability. These findings suggest that *vaccine delay* is shaped not only by access constraints, but also by how individuals perceive and respond to delayed protection.

From a behavioral medicine perspective, the results highlight the importance of service design in preventive health behavior. Reducing **waiting time**, improving the predictability of access, and strengthening institutional trust may help support more timely and equitable vaccination uptake. More broadly, these findings suggest that behavioral responses to delayed protection should be considered when designing preventive health programs and evaluating barriers to timely care.

Declaration of Generative AI Use

Generative artificial intelligence tools (including ChatGPT) were used to assist with language editing, formatting, and clarification of exposition during manuscript preparation. These tools were not used to generate data, conduct analyses, interpret results, or draw scientific conclusions. All analyses and

interpretations were performed by the authors, who take full responsibility for the content of the manuscript.

Ethics approval and informed consent

The study involved anonymous, minimal-risk survey research conducted in China. All study procedures complied with relevant national regulations and institutional guidelines. The study protocol was reviewed by an appropriate institutional ethics committee and determined to require no formal ethics approval. All participants provided informed consent prior to participation.

Data Availability Statement

The data that support the findings of this study are not publicly available due to ethical and privacy considerations related to human-subject survey data. De-identified data may be made available from the corresponding author upon reasonable request, subject to approval by the relevant ethics committee.

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Online Supplementary Material

How Waiting Time Shapes Preventive Health Behavior: Evidence from Vaccination Decisions

This document provides supplementary materials referenced in the main manuscript, including instrument materials and validity checks, sample benchmarking, model estimation details and derivations, subgroup-specific plots, trade-off computations, and an exploratory epidemic simulation.

Appendix A. DCE Instrument, Choice Tasks, and Validity Diagnostics

This appendix presents the diagnostic checks used to assess internal validity in the discrete choice experiment (DCE), including (i) dominance behavior, (ii) attribute non-attendance (ANA), (iii) transitivity and consistency, (iv) intra-individual variable correlations, and (v) pilot-study procedures. Collectively, these checks support the internal reliability and behavioral coherence of the responses.

A.1 Dominance Behavior

Dominance tests assess whether respondents ever chose an alternative that was strictly dominated on all attributes. As shown in Table 1, only 1.8% of respondents violated dominance at least once, a rate that is low by the standards commonly used in DCE research. This pattern suggests that most respondents understood the choice tasks and engaged in meaningful trade-offs.

A.2 Attribute Non-Attendance (ANA)

ANA analyses assess whether respondents systematically ignored particular attributes during choice tasks. Table 1 indicates that ANA was 0% for waiting time, efficacy, side effects, and cash incentives, reflecting strong salience of the primary decision attributes. A small minority (5.4%) showed insensitivity to vaccine origin, suggesting minor simplification but no evidence of widespread attribute non-attendance. Overall, ANA diagnostics demonstrate robust engagement with the full attribute structure.

Table 1: Dominance behavior and attribute non-attendance (ANA)

Check	**Definition**	**Result (%)**
Dominance rate	Share selecting a dominated alternative	1.8
ANA: Waiting time	Choices invariant to waiting-time levels	0.0
ANA: Vaccine efficacy	Choices invariant to efficacy levels	0.0
ANA: Side effects	Choices invariant to side-effect levels	0.0
ANA: Cash incentives	Choices invariant to cash levels	0.0
ANA: Vaccine origin	Choices invariant to origin levels	5.4

Note. Dominance reflects selecting an option strictly inferior across all attributes. ANA indicates choices unaffected by variation in a given attribute across tasks.

A.3 Transitivity and Choice Consistency

Respondents' choices satisfied the transitivity assumptions of random utility theory. No systematic violations or preference reversals were detected, and repeated patterns across the six tasks reflected stable and ordered preferences. These findings support the internal consistency and construct validity of the DCE, reinforcing the reliability of the resulting preference and time-discount parameter estimates.

A.4 Correlation Matrix of Intra-Individual Variables

Figure 1 displays pairwise correlations across all individual-level covariates included in the mixed-logit and subgroup models. All coefficients were below 0.10, indicating minimal concerns regarding multicollinearity or redundancy.

Access to healthcare showed only weak associations with other variables. Government trust was slightly positively correlated with age and rural residence, while smoking exhibited a weak negative correlation with chronic medical conditions. Overall, these low correlations confirm sufficient distinctness across covariates for use in the multivariate analyses.

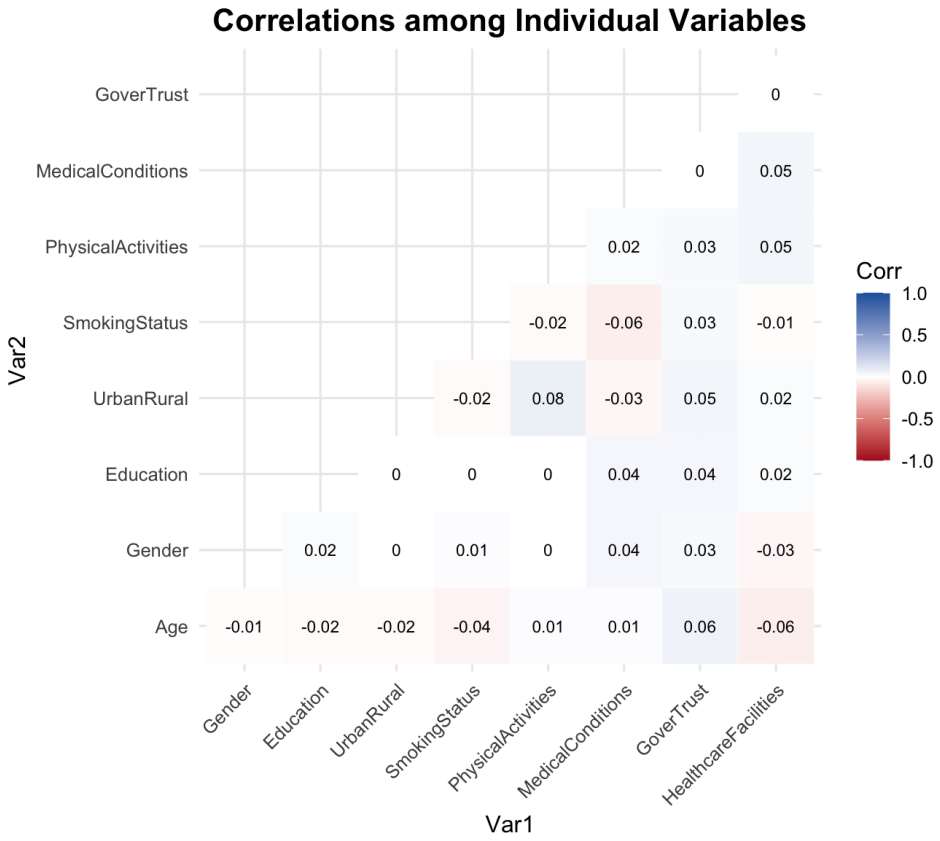


Figure 1: Correlation matrix of individual-level covariates. Darker colours indicate stronger correlations. All coefficients are below 0.10, confirming negligible multicollinearity.

A.5 Example Choice Task

Figure 2 provides an example of the choice task presented to respondents. Each task featured two hypothetical vaccine profiles varying across five attributes—waiting time, origin, efficacy, side-effect severity, and cash incentives—and an opt-out alternative. Attribute descriptions appeared at the top of the screen. For example, efficacy was defined behaviorally as the percentage reduction in infection risk ("95% effectiveness means a 95% reduction in infection risk"), and side effects were described using both text and images to reduce cognitive burden.

The example shown contrasts two vaccines identical in origin and efficacy but differing in waiting time (0 vs. 1 month), side-effect severity (fatigue for one week vs. fever lasting two days), and incentives (800 CNY vs. 200 CNY).

· 想象一下，您正在决定是否为未来的疫情爆发接种类似新冠病毒的疫苗。您有两种疫苗选择，具有以下特征，或者您可以选择不接种任何疫苗。根据提供的信息，您会选择哪个选项

疫苗效力以百分比表示，表示疾病风险降低（例如，95% 的有效性意味着风险降低 95%）。

疫苗平均等待时间	0（无需等待）	1 个月	我永远不会接种疫苗
疫苗产地	进口疫苗	进口疫苗	
疫苗功效	95 %	95 %	
副作用	重度（疲劳 1 周）	中度（发热 2 天）	
现金奖励	800 元	200 元	

上一题 1/6 下一题

Figure 2: Example DCE choice task shown to respondents. Participants selected between two hypothetical vaccine profiles or an opt-out option.

A.6 Pilot Study Procedures

A pilot study was conducted with 60 adults from Wuhan to assess the clarity, feasibility, and cognitive burden of the DCE instrument. The pilot followed the same screening criteria as the main study and included the full sequence of choice tasks, sociodemographic questions, and a debriefing module.

The pilot served three purposes. First, it evaluated comprehension of attribute descriptions and level wording. Respondents demonstrated clear understanding of waiting time, efficacy, and side-effect attributes, while minor revisions were made to simplify phrasing for vaccine origin and severity icons.

Second, the pilot assessed task difficulty and respondent burden. Completion times ranged from 4.7 to 10.2 minutes (mean: 6.1), indicating that the number of tasks and cognitive demands were appropriate. Dominance and ANA rates were consistent with those in the main sample, suggesting no evidence of satisficing or disengagement.

Third, the pilot refined the statistical efficiency of the experimental design. Preliminary parameter estimates informed updated priors in the efficient design algorithm, improving attribute balance and orthogonality in the final DCE.

Overall, pilot results confirmed that the survey instrument was clear, feasible, and capable of generating high-quality preference data.

Appendix B. Sample Representativeness and Census Benchmarking

This appendix evaluates the representativeness of the analytic sample ($N = 1{,}027$) by comparing key demographic characteristics with benchmarks from the 2020 Wuhan Census [**?**]. Sampling quotas were used to align the sample with the city's adult population on gender, age, and district of residence. As shown in Table 2, the sample closely approximates the demographic profile of Wuhan's adult population. Minor differences—such as a slightly higher

proportion of college-educated respondents—are typical of online survey panels and were further examined through robustness and weighting analyses.

Table 2: Comparison of analytic sample with 2020 Wuhan Census benchmarks

Characteristic	**Sample (%)**	**Wuhan Census (%)**
Gender		
Female	53.0	51.1
Male	47.0	48.9
Age distribution		
18–34	21.0	23.9
35–44	31.2	28.6
45–54	22.5	21.7
55–65	13.2	14.3
65+	12.0	11.5
Residence		
Urban	47.0	48.3
Rural	53.0	51.7
Education level		
No formal schooling	4.0	5.6
Primary school	22.0	24.3
Middle school	25.7	27.1
High school	20.4	21.7
College/University	15.6	13.4
Graduate degree	11.0	7.9

Note. Census benchmarks are drawn from the Wuhan Bureau of Statistics [**?**]. Deviations follow typical patterns in online survey panels. Post-stratification weighting yielded mixed-logit and discount-rate estimates substantively identical to unweighted results (Appendix C).

Summary of sample representativeness

- Survey sampling produced close alignment with Wuhan's 2020 Census for gender, age, and urban–rural residence.
- Slight overrepresentation of college-educated respondents is consistent with online panel sampling.
- Pearson χ^2 tests indicate no statistically significant difference between sample and population distributions at the 5% level.
- Weighted and unweighted mixed-logit models produced nearly identical results, confirming robustness to demographic composition (see Appendix C).

To formally evaluate representativeness, Pearson χ^2 tests were conducted for gender, age group, and residence. All tests were statistically non-significant at the 5% level, indicating no systematic deviation from Wuhan's adult population. Weighted mixed-logit models yielded parameters nearly identical to those in the unweighted sample, reinforcing the robustness of the main findings.

Appendix C. Model Estimation Details

This appendix provides additional detail on mixed-logit estimation and the derivation of exponential and hyperbolic discount parameters used in the main analysis. Throughout, the experimentally manipulated attribute is referred to as **waiting time**, while *vaccine delay* refers to the corresponding behavioral consequence in real-world uptake.

C1. Utility Specification

We adopt a random-utility framework in which individual i evaluates vaccine alternative j in choice task t according to:

$$U_{ijt} = \beta_0 + \beta_1 \,\text{WaitingTime}_{\text{std},ijt} + \beta_2 \,\text{VaccineOrigin}_{\text{std},ijt} + \beta_3 \,\text{VaccineEfficacy}_{\text{std},ijt} + \beta_4 \,\text{SideEffects}_{\text{std},ijt} + \beta_5 \,\text{CashIncentives}_{\text{std},ijt} + \varepsilon_{ijt}, \tag{C1}$$

where ε_{ijt} follows an i.i.d. Type I extreme-value distribution. For comparison, both conditional logit (MNL) and mixed logit (MXL) models were estimated, but the MXL specification is preferred because it allows random coefficients to capture unobserved preference heterogeneity.

In vector notation:

$$U_{ijt} = \beta_0 + \sum_{k=1}^{K} \beta_k X_{k,ijt} + \varepsilon_{ijt}. \tag{C2}$$

Standardization of attribute coefficients facilitates comparison across attributes and feeds directly into the discount-rate estimation procedure described below.

C2. Discount Function Specification

Observed waiting-time levels $t \in \{0, 1, 3, 6\}$ months are mapped to standard exponential and hyperbolic discounting frameworks:

$$D_{\text{Exp}}(t; \hat{\delta}) = e^{-\hat{\delta} t}, \qquad \hat{\delta} \geq 0, \tag{C3}$$

$$D_{\text{Hyper}}(t; \hat{\kappa}) = \frac{1}{1 + \hat{\kappa} t}, \qquad \hat{\kappa} \geq 0. \tag{C4}$$

These formulations correspond to time-consistent (exponential) and present-biased (hyperbolic) discounting, respectively.

C3. Constructing Observed Discount Points

Let $\beta_{\text{waiting time (per month)}}$ be the unstandardized coefficient from the secondary model in which waiting time is entered linearly. This coefficient represents the marginal utility loss associated with a one-month increase in waiting time and is recovered from the standardized coefficient via:

$$\beta_{\text{waiting time (per month)}} = \frac{\beta_{\text{waiting time (std)}}}{\text{SD(MultipleDelays)}}. \tag{C5}$$

Following standard practice in preventive-health time-preference studies, we define the observed discount factor for each delay t as:

$$D_{\text{obs}}(t) = \exp\left(-\beta_{\text{waiting time (per month)}}\, t\right). \tag{C6}$$

These points serve as empirical inputs for fitting exponential and hyperbolic discount functions.

C4. Estimation of Discount Parameters

The exponential and hyperbolic discount parameters are obtained by minimizing the sum of squared errors (SSE) between the observed and theoretical discount functions.

Hyperbolic fit

$$\text{SSE}_{\text{Hyper}}(\hat{\kappa}) = \sum_{i=1}^{n} \left[D_{\text{obs}}(t_i) - D_{\text{Hyper}}(t_i; \hat{\kappa})\right]^2. \tag{C7}$$

The first-order condition is:

$$\frac{\partial}{\partial \kappa}\mathrm{SSE}_{\mathrm{Hyper}}(\kappa) = 2\sum_{i=1}^{n}\left(D_{\mathrm{obs}}(t_i) - \frac{1}{1+\kappa t_i}\right)\frac{t_i}{(1+\kappa t_i)^2} = 0, \qquad \text{evaluated at } \kappa = \hat{\kappa}. \quad \text{(C8)}$$

Exponential fit

$$\mathrm{SSE}_{\mathrm{Exp}}(\hat{\delta}) = \sum_{i=1}^{n}\left[D_{\mathrm{obs}}(t_i) - D_{\mathrm{Exp}}(t_i;\hat{\delta})\right]^2. \quad \text{(C9)}$$

First-order condition:

$$\frac{\partial}{\partial \delta}\mathrm{SSE}_{\mathrm{Exp}}(\delta) = 2\sum_{i=1}^{n}\left(D_{\mathrm{obs}}(t_i) - e^{-\delta t_i}\right)t_i e^{-\delta t_i} = 0, \qquad \text{evaluated at } \delta = \hat{\delta}. \quad \text{(C10)}$$

Closed-form solutions do not exist; parameters were estimated via nonlinear least squares.

C5. Subgroup Estimation and Multiplicity

Subgroup-specific MXL models were estimated separately for demographic, behavioral, health-status, and institutional-trust categories. Because this involves multiple hypothesis tests, p-values for subgroup waiting-time coefficients were adjusted using the Benjamini–Hochberg false-discovery rate (FDR) at $q = 0.10$. Robustness of subgroup differences was confirmed under Holm–Bonferroni corrections.

C6. Recovering Discount Parameters from Waiting-Time Coefficients

To recover exponential discount parameters from the DCE model, we map the estimated waiting-time coefficient $\beta_{\text{waiting time}}$ to the implied monthly discount factor δ. Under exponential discounting, utility declines with delay according to $U(t) = U_0\delta^t$. A first-order approximation yields a linear representation in which the marginal disutility of delay corresponds to the logarithm of the discount factor. Normalizing baseline utility to $U_0 = 1$

implies:

$$\hat{\delta} = \exp\left(\hat{\beta}_{\text{waiting time}}\right).$$

Thus, more negative values of $\beta_{\text{waiting time}}$ correspond to lower discount factors and stronger impatience toward delayed protection.

C7. MWTA Derivation

The marginal willingness-to-accept (MWTA) for one month of waiting time is derived as the ratio of the waiting-time coefficient to the cash-incentive coefficient:

$$\text{MWTA} = -\frac{\beta_{\text{waiting time}}}{\beta_{\text{cash}}}.$$

This represents the monetary compensation required to offset the disutility of an additional month of waiting.

C8. Additional Diagnostics

We conducted additional diagnostic checks, including model comparison (AIC and log-likelihood), scale-heterogeneity assessment using a GMNL specification, and re-estimation excluding potential speeders. Across all alternative specifications, the sign, magnitude, and statistical significance of the waiting-time coefficient remained highly stable, and the implied discount parameters were unchanged. These results confirm that the main findings are robust to functional-form assumptions and response-quality concerns.

C9. Behavioral Delay Multiplier (BDM)

To provide an intuitive interpretation of the hyperbolic discount parameter, we compute a Behavioral Delay Multiplier (BDM), defined as:

$$\text{BDM} = \frac{1}{1 - \hat{\kappa}}, \tag{C11}$$

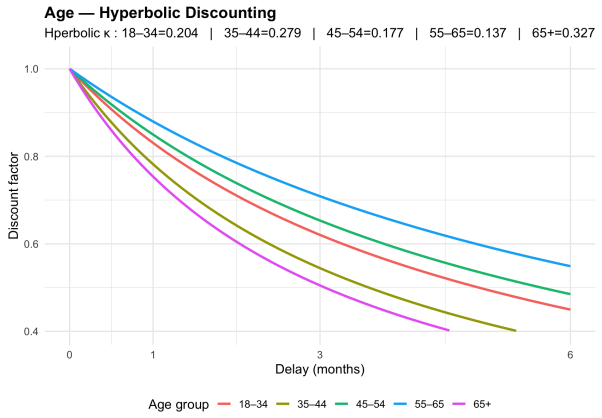
which rescales the estimated hyperbolic parameter into a measure of the perceived cost of waiting. A value greater than 1 indicates that each month of objective waiting is experienced as a longer behavioral delay. For example, the overall estimate $\hat{\kappa} = 0.225$ implies a BDM of 1.29, meaning that one month of waiting feels equivalent to approximately 1.29 months. Subgroups with larger $\hat{\kappa}$—such as older adults, respondents with primary education, or those reporting neutral trust in government—exhibit BDM values between 1.45 and 1.55, indicating substantially amplified perceived waiting costs. In contrast, groups with smaller $\hat{\kappa}$, including rural residents and individuals reporting regular physical activity, show BDM values near 1.15–1.20, reflecting weaker behavioral sensitivity to delay.

Table 3: Behavioral Delay Multiplier (BDM) for selected subgroups

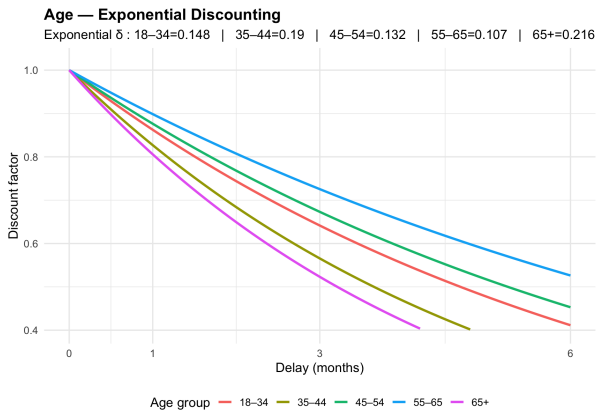
Subgroup	$\hat{\kappa}$	**BDM**	**Interpretation**
Overall sample	0.225	1.29	1 month feels like 1.29 months
Urban residents	0.290	1.41	Higher perceived waiting cost
Rural residents	0.174	1.21	Lower perceived waiting cost
Primary education	0.348	1.53	Amplified behavioral delay
Adults aged 65+	0.327	1.49	Delay feels substantially longer
Neutral trust in gov.	0.356	1.55	Strongest impatience

Appendix D. Subgroup-Specific Discounting Plots

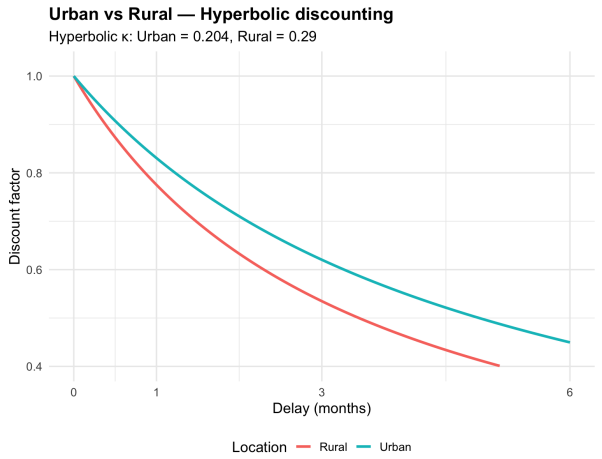
This appendix presents subgroup-specific discounting profiles to complement the summary heterogeneity results reported in the main text.



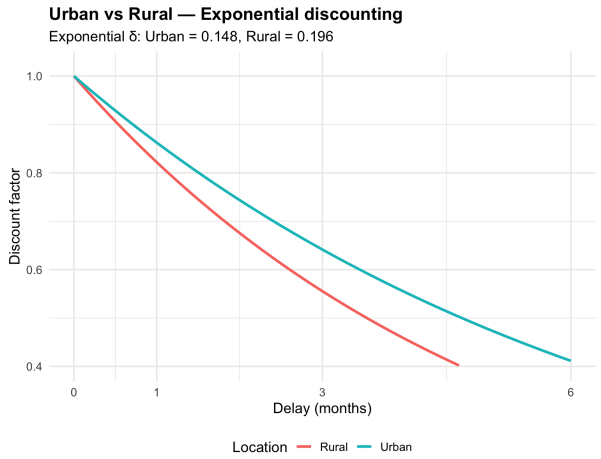
Panel A: Age-specific discounting. Five age groups (18–34, 35–44, 45–54, 55–65, 65+).



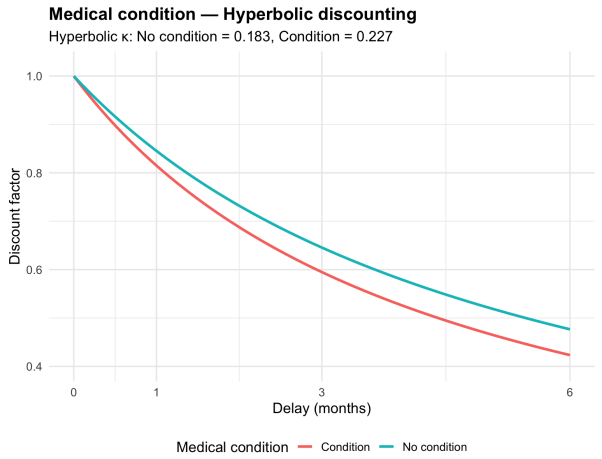
Five age groups (18–34, 35–44, 45–54, 55–65, 65+).



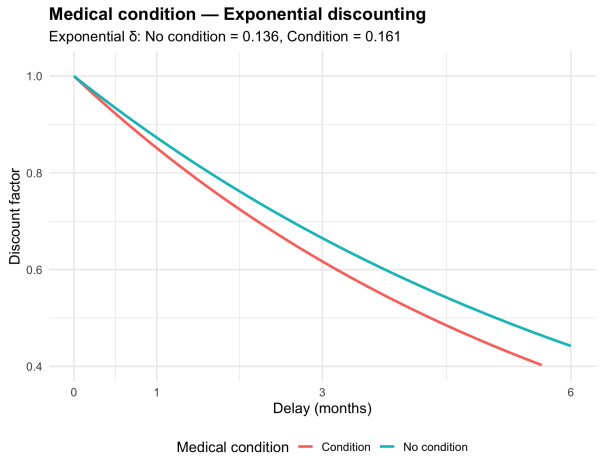
Panel B: Residence-specific discounting.



Urban versus rural respondents.

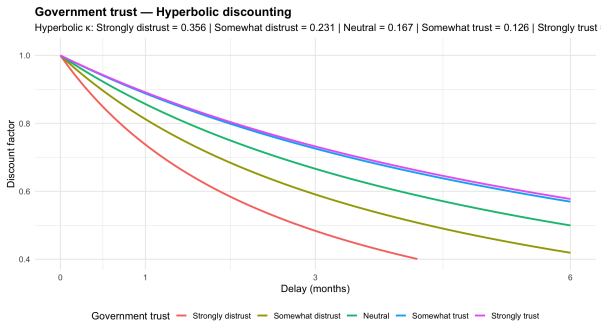


Panel C: Medical-condition-specific discounting.

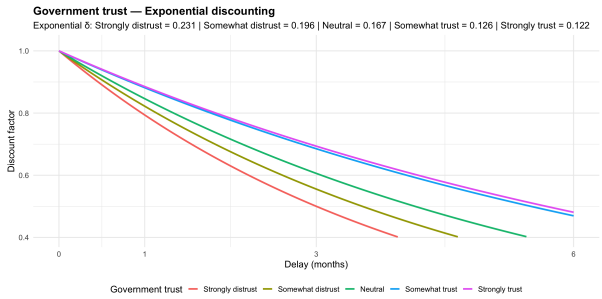


Individuals with versus without medical conditions.

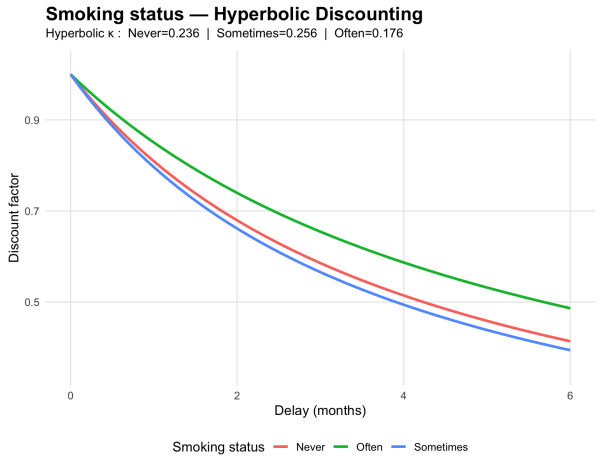
Figure 3: **Appendix Figure D1.** Subgroup-specific discounting (Panels A–C). Each panel compares observed discounting with hyperbolic and exponential fits.



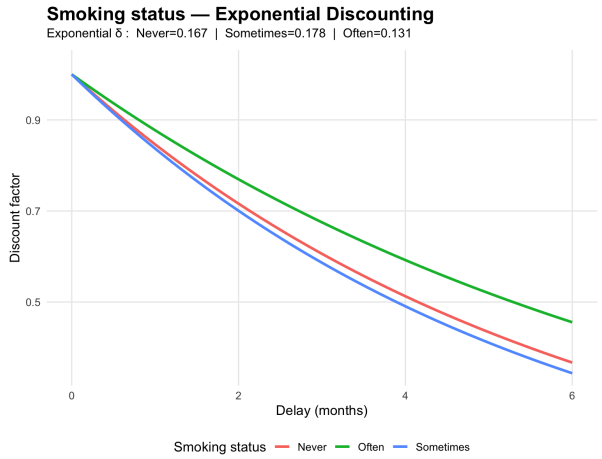
Panel D: Government-trust-specific discounting.



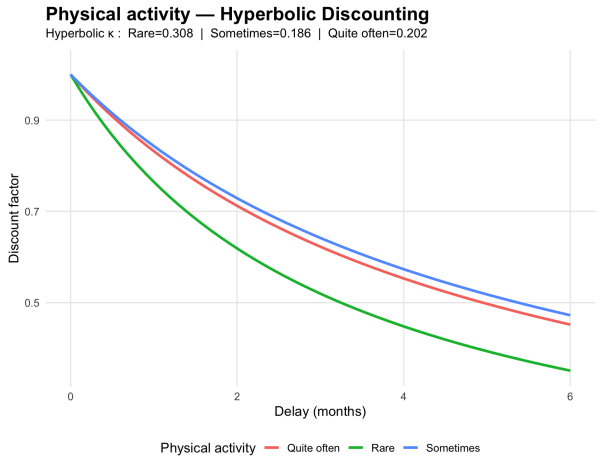
Five levels of government trust.



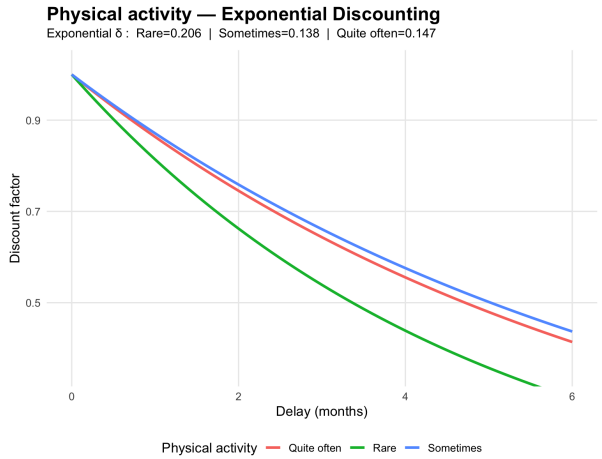
Panel E1: Smoking-status-specific discounting.



Never, sometimes, often.



Panel E2: Physical-activity-specific discounting.



Rare, sometimes, quite often.

Figure 4: **Appendix Figure D2.** Subgroup-specific discounting (Panels D–E). Each row shows hyperbolic versus exponential discount functions for a subgroup category.

Appendix E. Trade-Off Computation between Waiting Time and Monetary Incentive

This appendix summarizes the trade-off between **waiting time** and cash incentives implied by the mixed-logit estimates.

The marginal rate of substitution (MRS) between **waiting time** and cash incentive was computed as

$$\mathrm{MRS}_{\text{waiting time–cash}} = -\frac{\beta_{\text{cash}}}{\beta_{\text{waiting time}}}. \tag{1}$$

This quantity summarizes the relative trade-off between monetary incentives and **waiting time**, holding other vaccine attributes constant.

Using the relevant mixed-logit coefficients, we obtained

$$\mathrm{MRS}_{\text{waiting time–cash}} = -\frac{0.102}{-0.125} = 0.82. \tag{2}$$

Under the coding used in the DCE, this estimate indicates that a 100 RMB increase in cash incentive offsets approximately 0.82 units of **waiting time**. Interpreted in relation to the waiting-time scale used in the experiment, this implies that respondents viewed a reduction of approximately one month in **waiting time** as behaviorally comparable to a modest monetary incentive. Confidence intervals were estimated using the delta method.

Appendix F. Exploratory Epidemic Simulation (SEIR Extension)

This appendix presents an exploratory population-level extension linking behavioral responses to **waiting time** to epidemic dynamics. The purpose of this extension is illustrative: it shows how heterogeneity in responses to delayed protection may affect the timing of effective vaccine uptake and, in turn, outbreak trajectories. This appendix is not intended as a

full forecasting exercise or a complete policy evaluation.

Model structure

We used a behaviorally augmented susceptible–exposed–infectious–removed (SEIR) framework. Let S_t, E_t, I_t, and R_t denote the susceptible, exposed, infectious, and removed populations at time t, with total population size $N = S_t + E_t + I_t + R_t$. Vaccination reduces the susceptible population through an effective vaccination rate that depends on both delivery capacity and behavioral acceptance. The model is given by

$$\frac{dS_t}{dt} = -\beta \frac{S_t I_t}{N} - v_{\text{eff}}(t) S_t,$$

$$\frac{dE_t}{dt} = \beta \frac{S_t I_t}{N} - \sigma E_t,$$

$$\frac{dI_t}{dt} = \sigma E_t - \gamma I_t,$$

$$\frac{dR_t}{dt} = \gamma I_t + v_{\text{eff}}(t) S_t,$$

where β is the transmission rate, σ is the progression rate from exposed to infectious status, and γ is the recovery rate.

Behavioral vaccination function

To incorporate behavioral responses to delayed protection, the effective vaccination rate is defined as

$$v_{\text{eff}}(t) = v_{\text{cap}}(t) \times P_t,$$

where $v_{\text{cap}}(t)$ denotes the maximum operational vaccination capacity at time t, and P_t denotes the probability of vaccine uptake implied by the estimated choice model under the relevant **waiting time** scenario.

In the benchmark scenario, uptake is constrained only by delivery capacity, such that $P_t = 1$. In the behavioral-response scenarios, $P_t < 1$ and varies according to the estimated delay sensitivity recovered from the discrete choice experiment. For subgroup-specific simulations, P_t was adjusted using subgroup-specific response parameters so that greater sensitivity to **waiting time** translated into slower effective uptake.

Calibration and simulation scenarios

The simulation was calibrated to illustrate how behavioral responses to **waiting time** may shift vaccination later in the epidemic curve. Epidemiological parameters were chosen to generate plausible outbreak dynamics for a COVID-like respiratory infection, while the behavioral component was informed by the estimated discounting parameters and subgroup-specific delay sensitivity reported in the main analysis.

We compared three classes of scenarios:

1. **Supply-only benchmark:** vaccine uptake is determined entirely by operational capacity, with no behavioral penalty from **waiting time**.

2. **Average behavioral-response scenario:** effective uptake is reduced according to the full-sample response to delayed protection.

3. **Subgroup behavioral-response scenarios:** effective uptake varies according to subgroup-specific sensitivity to **waiting time**, allowing illustration of heterogeneity in epidemic consequences.

The primary outcome was the cumulative attack rate. We also examined representative infection trajectories to illustrate how behavioral responses to delayed protection may alter the timing and magnitude of infections over the course of an outbreak.

Interpretation

This extension should be interpreted as a behavioral illustration rather than a structural epidemic forecast. The goal is to demonstrate that **waiting time**, estimated at the individual level as a behavioral barrier to vaccination, may also have population-level implications when it slows effective uptake during the early growth phase of an outbreak. In this sense, *vaccine delay* can arise not only from logistical constraints but also from behavioral responses to delayed protection.

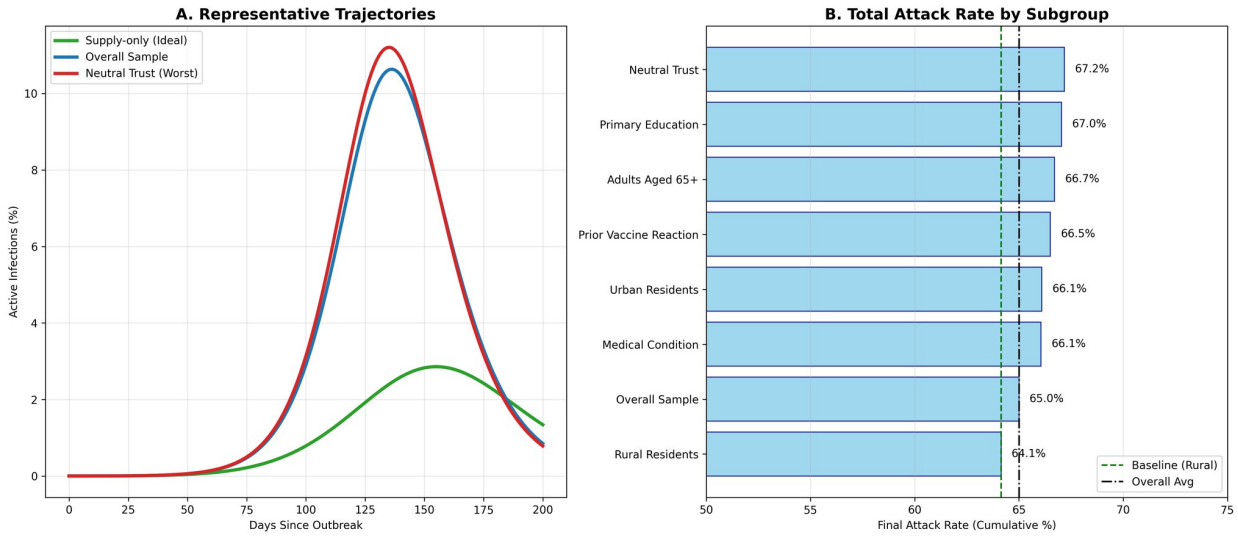


Figure 5: **Appendix Figure F1.** Exploratory SEIR-based simulation. **Panel A** shows representative infection trajectories under a supply-only benchmark and under behavioral-response scenarios. **Panel B** summarizes cumulative attack rates across selected subgroups, illustrating how stronger sensitivity to delayed protection can shift effective uptake later in the outbreak and increase cumulative infections.

Note. This SEIR extension is included to illustrate how estimated behavioral responses to **waiting time** may translate into population-level epidemic consequences. Full scenario assumptions, additional robustness checks, and further calibration details can be provided if needed.